

		$r = 3 \text{ } \Omega$
18	C	$R = \frac{\rho L}{A}$ Since volume of wire stays constant, $V = AL = A_1(4L)$ $A_1 = \frac{A}{4}$ $R_1 = \frac{\rho L_1}{A_1} = \frac{16\rho L}{A} = 16R$
19	D	Applying Ohm's law